IP ADDRESSING & SUBNETTING

Course: Diploma in Cyber Security
Topic: IP Addressing & Subnetting
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Introduction

IP addressing gives devices an address so they can communicate on a network. Subnetting divides a larger network into smaller networks. In this assignment, I have covered IP address classes, basic subnet calculations and simple subnetting examples.

Task 1: IP Address Classes

Class	General Range	Main Use	Example
A	1.0.0.0 – 126.255.255.255	Very large networks	Large organization network
B	128.0.0.0 – 191.255.255.255	Medium-sized networks	University or company network
C	192.0.0.0 – 223.255.255.255	Smaller networks	Home/office private network such as 192.168.1.0/24

D	**224.0.0.0 – 239.255.255.255**	**Multicast**	**Video or live-stream multicast traffic**
E	**240.0.0.0 – 255.255.255.255**	**Reserved / experimental**	**Research and testing**

Note: These are the traditional IPv4 class ranges. Modern networks mainly use CIDR instead of classful addressing.

Task 1: IPv4 Address Classes

Class	General Range	Main Use	Example
A	1.0.0.0 – 126.255.255.255	Very large networks	Large organization network
B	128.0.0.0 – 191.255.255.255	Medium-sized networks	University or company network
C	192.0.0.0 – 223.255.255.255	Smaller networks	Home/office private network 192.168.1.0/24
D	224.0.0.0 – 239.255.255.255	Multicast	Video/live-stream multicast traffic
E	240.0.0.0 – 255.255.255.255	Reserved / experimental	Research and testing

Task 2: Calculate Network, Broadcast and Host Range

Given IP address: 192.168.10.45
Subnet mask: 255.255.255.0 (/24)

Step 1: A /24 mask means the first three octets are the network part, so the network address is 192.168.10.0.

Step 2: Set all host bits to 1. The broadcast address becomes 192.168.10.255.

Step 3: The first usable host is 192.168.10.1 and the last usable host is 192.168.10.254.

Final Answer: Network Address = 192.168.10.0; Broadcast Address = 192.168.10.255; Valid Host Range = 192.168.10.1 – 192.168.10.254.

Task 2: IPv4 Subnetting & Host Calculation

Given IP 192.168.10.45 with Subnet Mask 255.255.255.0 (/24)

Network Address	192.168.10.0
First Usable Host	192.168.10.1
Last Usable Host	192.168.10.254
Valid Host Range	192.168.10.1 – 192.168.10.254
Broadcast Address	192.168.10.255

Task 3: Four-Subnet Plan for a Food Delivery Startup

Starting network: 10.0.0.0/24. To create 4 equal subnets, I borrow 2 host bits because $2^2 = 4$. The new prefix is /26 and the subnet mask is 255.255.255.192.

Group	Network Address	Subnet Mask	Usable Host Range
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254

Each /26 subnet has 64 total addresses and 62 usable host addresses.

Task 4: College Wi-Fi with at Least 20 Subnets

Given network: 172.16.0.0/16.

Step 1: $2^4 = 16$, which is not enough. $2^5 = 32$, so I borrow 5 bits.

Step 2: New prefix = /16 + 5 = /21.

Step 3: /21 subnet mask = 255.255.248.0.

Step 4: Host bits left = 32 - 21 = 11. Total addresses per subnet = $2^{11} = 2048$. Usable hosts = 2048 - 2 = 2046.

Final Answer: New subnet mask = 255.255.248.0 (/21). Number of subnets = 32. Usable hosts per subnet = 2046.

Task 5: Invalid IP Address 192.168.1.300

The error is that each IPv4 number must be from 0 to 255. The value 300 is outside this range, so 192.168.1.300 is

not a valid IP address. For a normal 192.168.1.0/24 subnet, a valid host could look like 192.168.1.25 or 192.168.1.100. The usual usable range is 192.168.1.1 to 192.168.1.254.

Conclusion

In this assignment, I learned how IP addresses are divided into classes and how subnet masks are used to find network and host ranges. I also learned how to divide a network into smaller subnets and identify an invalid IP address.

Quick Revision

- **IPv4 octet values go from 0 to 255.**
- **/24 = 255.255.255.0.**
- **A /26 network gives 4 equal subnets from a /24.**
- **A /21 network has 2046 usable hosts per subnet.**
- **Network and broadcast addresses are not normally assigned to hosts.**